# Homework 10

*Due: Friday, April 29*

All homeworks are due at 11:59 PM on Gradescope.

**Please do not include any identifying information about yourself in the handin, including your Banner ID.**

Be sure to fully explain your reasoning and show all work for full credit.

**LaTeX tips**

Here are some LaTeX commands that might help you

`\overline{G}` $\overline{G}$

LaTeX *is* required. You should be using the `cs22` document class. Look for a LaTeX guide here for more information.

## Problem 1

$\mathbb{E}$ is the edge set of a complete graph with vertices $V$. Consider a graph $G$, where $V(G) = V$ and $E(G) \subseteq \mathbb{E}$. The complement of $G$, $\overline{G}$, is the graph where $V(\overline{G}) = V$ and $E(\overline{G}) = \mathbb{E} \setminus E(G)$.

Prove that, if $G$ is not connected, then $\overline{G}$ is connected.

# Problem 2

An orientation of a simple graph $G$ is an assignment which replaces each edge $\{u, v\} \in E(G)$ with one of the *ordered pairs* $(u, v)$ or $(v, u)$ but not both.

An orientation is *acyclic* if it has no directed cycles. That is, we could not have the following edges: $(x_1, x_2), (x_2, x_3), \ldots, (x_{k-1}, x_k), (x_k, x_1)$.

An orientation is *transitive* if whenever $(u, v) \in E(G)$ and $(v, w) \in E(G)$, then $(u, w) \in E(G)$.

a. Prove that a transitive orientation is necessarily acyclic.

b. Prove that there exists a transitive orientation of every bipartite graph.[1]

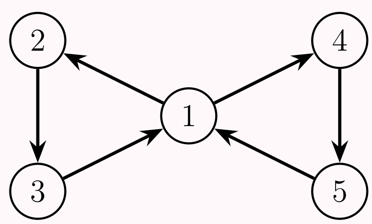
c. Suppose we randomly orient the complete graph $K_n$ by choosing an orientation for each edge equally likely (with probability $\frac{1}{2}$). What is the expected number of simple directed cycles? Give your answer as a mathematical expression in terms of $n$.

Updated Apr 26

Example Added Apr 26

**Example**

In the graph below:




The cycle $1 \to 2 \to 3 \to 1 \to 4 \to 5 \to 1$ is *not* a simple cycle, as the 1 vertex is repeated aside from the first and last position. This cycle, however, is comprised of simple cycles $1 \to 2 \to 3 \to 1$ and $1 \to 4 \to 5 \to 1$.

Additionally, the starting point of the cycle *does* matter: $1 \to 2 \to 3 \to 1$ is *not* the same cycle as $2 \to 3 \to 1 \to 2$.

*(Due to prior ambiguity, we **will** be accepting alternate solutions where the starting point of the cycle **doesn't** matter, for full credit.)*

[1] A *bipartite graph* is a graph that can be partitioned into two sets of vertices, $U$ and $V$, such that every edge connects a vertex in $U$ with a vertex in $V$. (Visually, this means that you can move some vertices to one side of the page and the rest of the vertices to the other side, and every edge will cross from one side of the page to the other.)

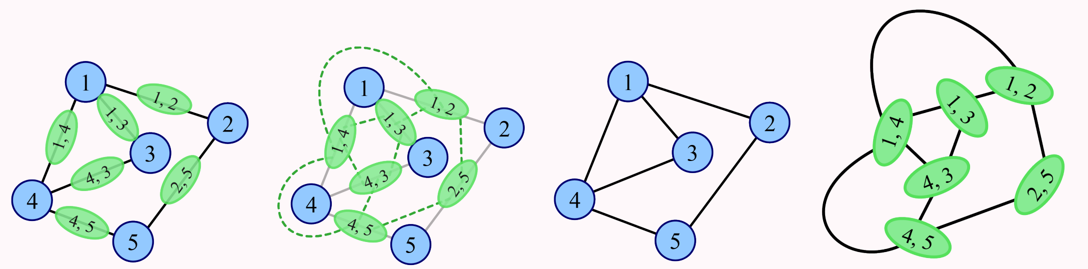
# Problem 3

Let $G$ be a graph with vertex set $V(G)$ and edge set $E(G)$. We construct the *line graph* of $G$, denoted by $L$, in the following way:

- For each edge in $E(G)$, there is exactly one corresponding vertex of $L$. That is, $V(L) = E(G)$.
- Two vertices of $L$ are connected by an edge if their corresponding edges in $G$ share a vertex.

**Example**

For example, here is a graph on the left with edges marked in green, and its corresponding line graph on the right where edges have become the nodes:[a]




[a] https://en.wikipedia.org/wiki/Line_graph

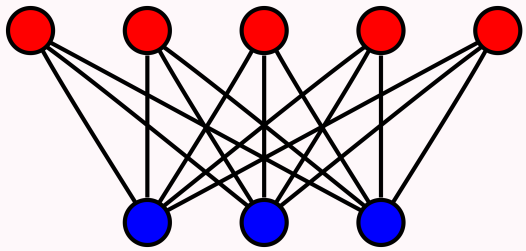
a. If $G$ has an Eulerian cycle, does $L$ necessarily have a Hamiltonian cycle? If yes, prove it. If no, give a counterexample.

*Problem 3 continued on next page...*

The *complete bipartite graph* $B_{m,n}$ is a graph with two sets of vertices $A$ and $B$ with $|A| = m$, $|B| = n$, an edge $\{a, b\}$ between every pair of vertices $a \in A$ and $b \in B$, and no edge between any pair of vertices from the same set.

**Example**

For example, here is $B_{5,3}$:[a]



[a] https://en.wikipedia.org/wiki/Bipartite_graph

b. Prove that in the line graph of any complete bipartite graph, every vertex has the same degree.

# ☕ Problem 4 (Mind Bender — *Extra Credit*)

*Mind Benders* are extra credit problems intended to be more challenging than usual homework problems and are an exploration into a topic not covered in lecture.

You might be familiar with *Minimum Spanning Trees* introduced in a previous course. We prove the correctness of *Kruskal's Algorithm*, which generates minimum spanning trees.

A graph $G$ is said to be *weighted* if there is a function $w : E(G) \to \mathbb{R}$. For any edge $e$, $w(e)$ is the weight of the edge.

A tree $T$ is said to be a *spanning tree* of $G$ if $V(T) = V(G)$ and $E(T) \subseteq E(G)$, that is, $T$ is a subgraph of $G$. Often, for a weighted graph, we want to find a minimum spanning tree (MST). $T$ is an MST of a graph $G$ if it is a[2] spanning tree that minimizes

$$\sum_{e \in E(T)} w(e).$$

Consider Kruskal's Algorithm, an algorithm for finding minimum spanning trees:

- Let $S = E(G)$, $E(T) = \emptyset$
- While $S$ is non-empty, remove a minimum weight edge from $S$. Add that edge to $E(T)$ if doing so will not create a cycle in $T$.

Prove that, if $G$ is connected, $T$ as constructed by Kruskal's algorithm is a minimum spanning tree. To do so, you must first prove $T$ is a spanning tree, then that it is minimal.

[2] A graph could have multiple MSTs! What if the weight of each edge is the same?